

LOAN DEFAULT & CREDIT RISK ANALYSIS

Project objective

Analyze customer demographics, income, credit score, loan characteristics, debt-to-income ratio and repayment/default patterns.

Dataset

Synthetic academic dataset created specifically for this project using the assignment's suggested columns. Records: 6,000; Loans: 6,000.

Key KPIs

Total customers: 6,000

Total loans: 6,000

Defaulted loans: 994

Non-defaulted loans: 5,006

Overall default rate: 16.57%

Average loan amount: 221,700.82

Average credit score: 689.75

Average annual income: 60,213.45

Average interest rate: 15.11%

Average debt-to-income ratio: 27.61%

Cleaning

Missing values were checked and filled appropriately; duplicates were checked and removed; numerical fields were converted to numeric types; credit-score and income groups were created.

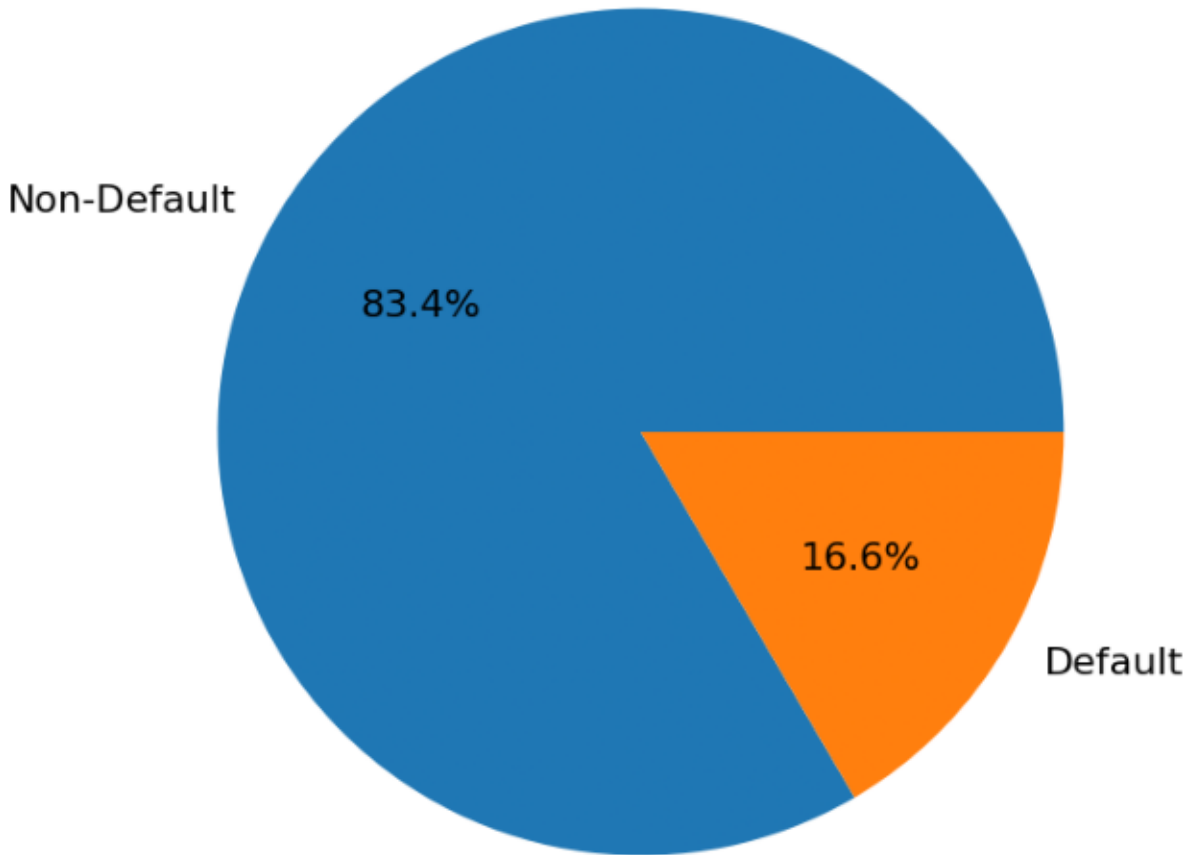
Business insights

1. Highest-default loan type: Personal.
2. Highest-default region: North.
3. Credit-score segment with highest default: Poor (<550).
4. Income segment with highest default: Low.
5. Employment segment with highest default: Unemployed.
6. Average credit score differs between defaulted and non-defaulted loans: 669.7 vs 693.7.
7. Average income differs between defaulted and non-defaulted loans: 53,452 vs 61,556.
8. Defaulted customers show a different average DTI: 35.9% vs 26.0%.
9. Previous defaults are an important risk indicator.
10. Credit score, DTI, income, loan amount and previous defaults should be monitored in credit-risk decisions.

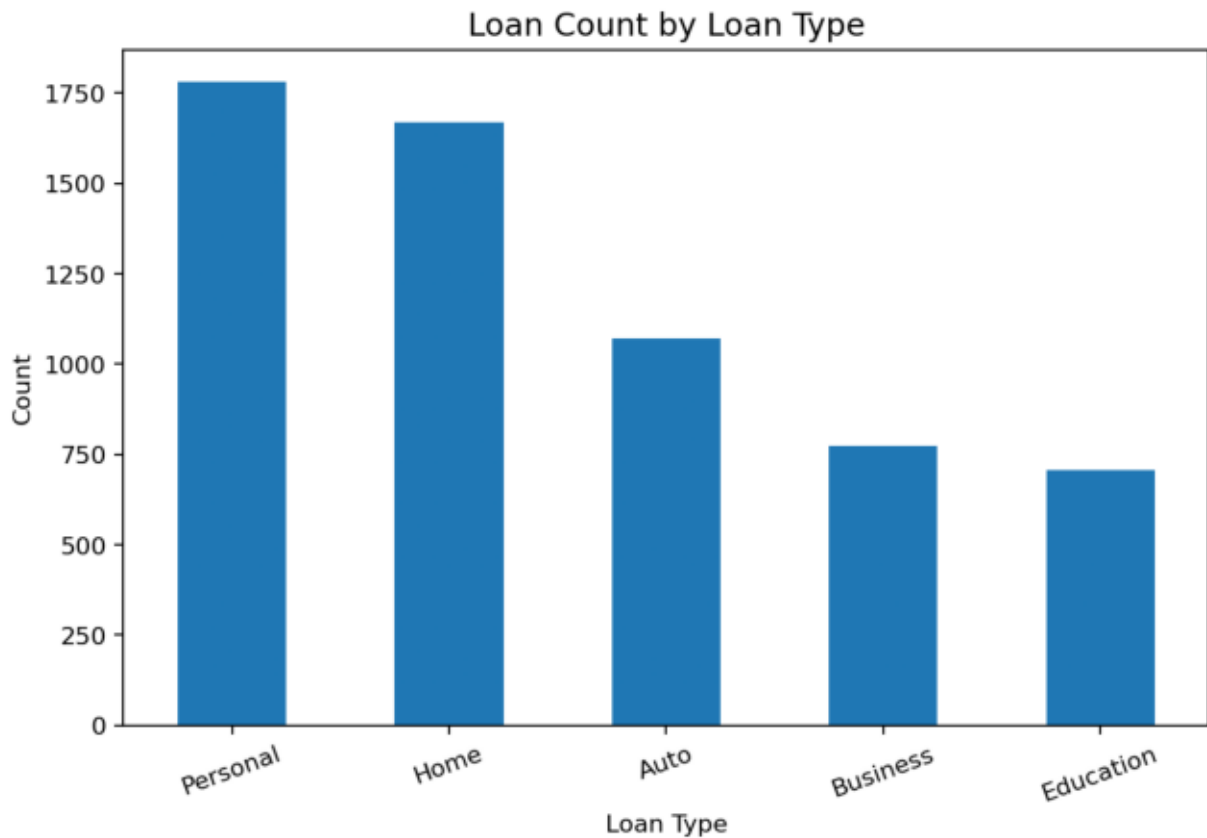
Conclusion

The analysis demonstrates how Python and Power BI can be used to identify loan-default patterns and support credit-risk monitoring.

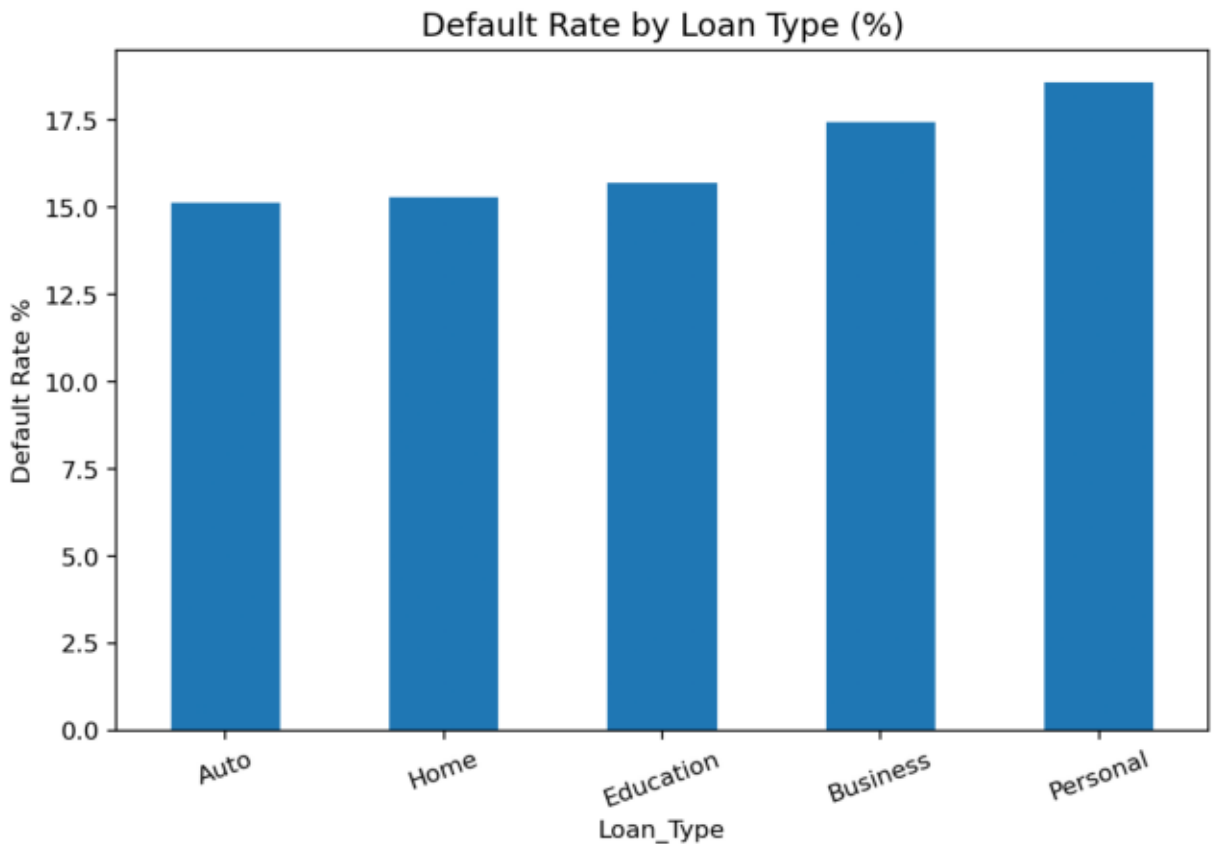
Loan Default Distribution



02 Loan Count by Loan Type

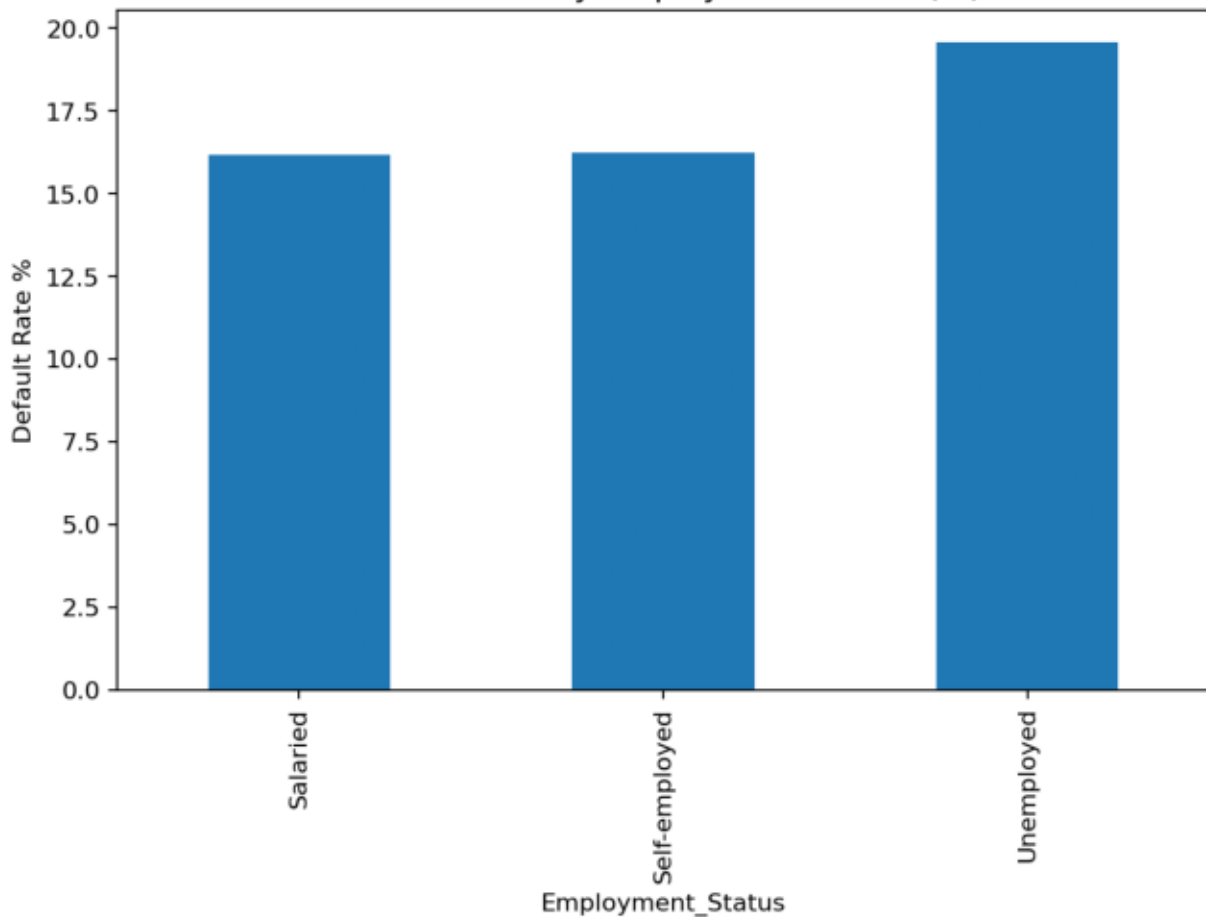


03 Default Rate by Loan Type

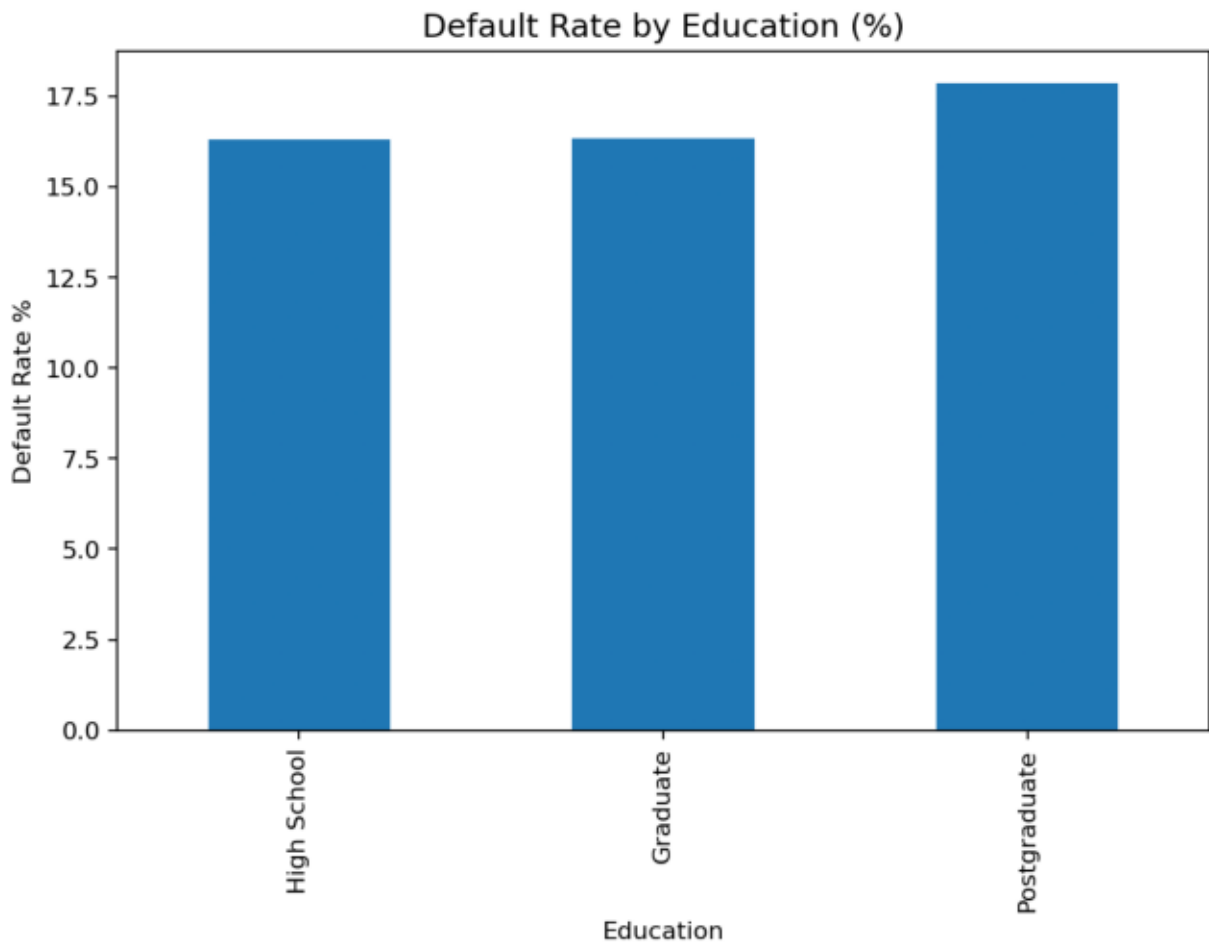


04 Default Rate by Employment

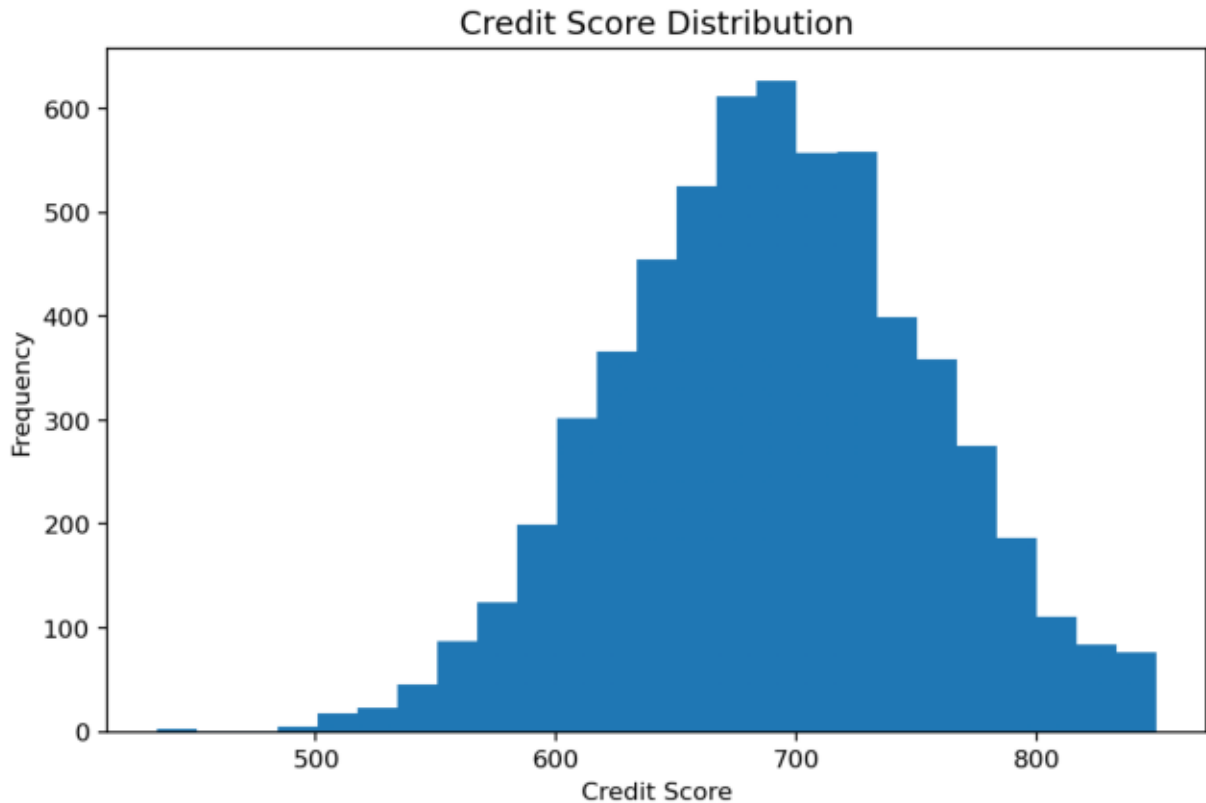
Default Rate by Employment Status (%)



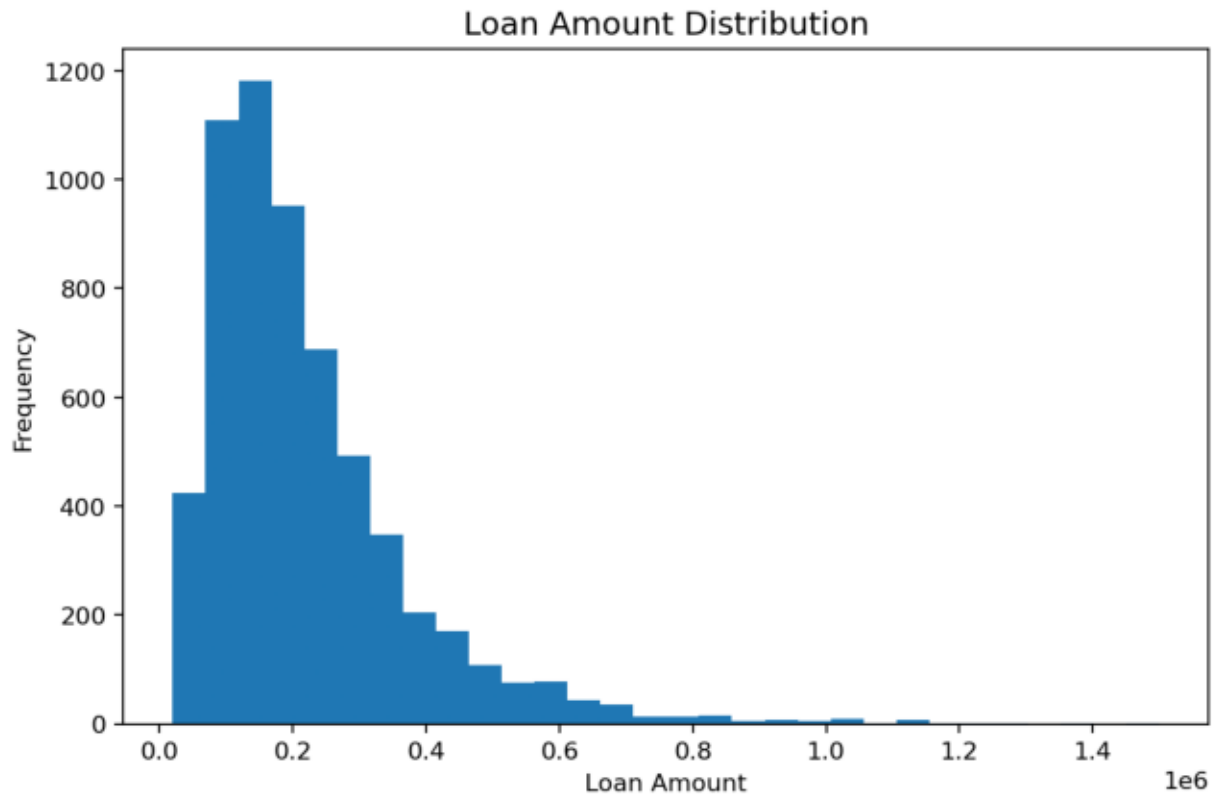
05 Default Rate by Education



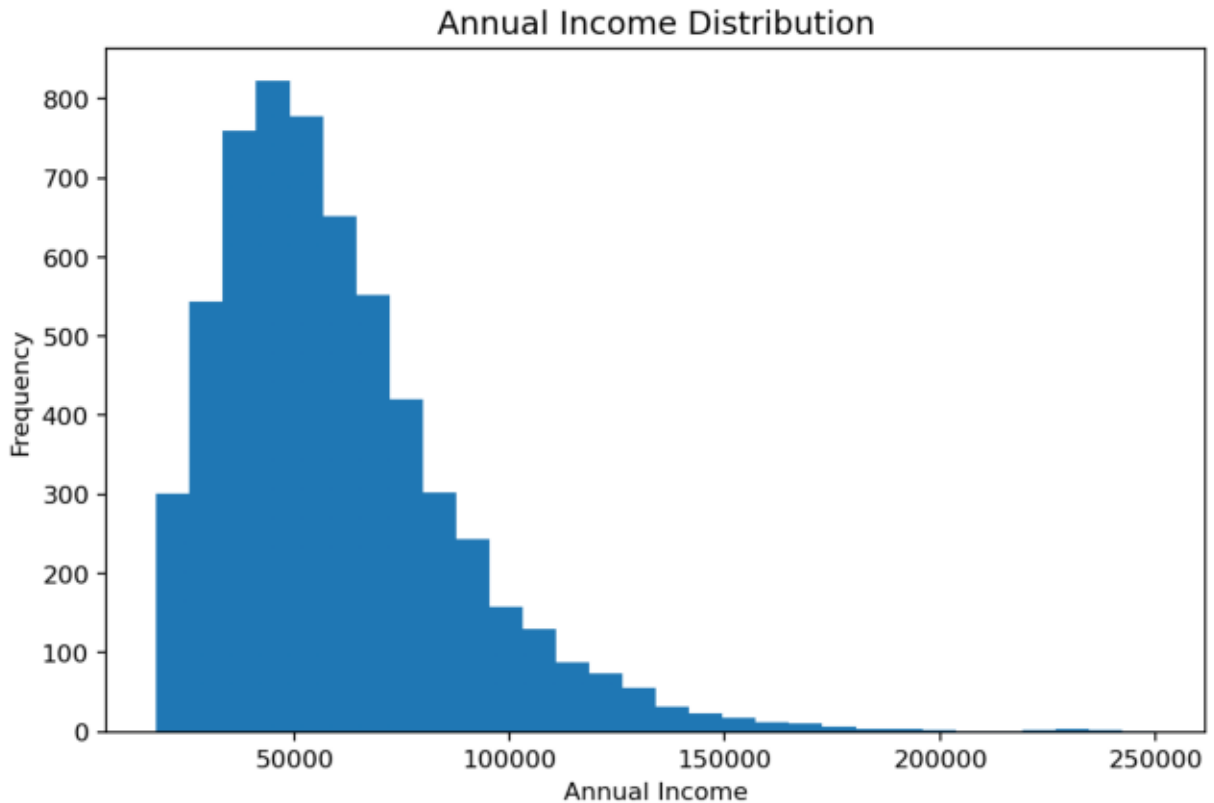
06 Credit Score Distribution



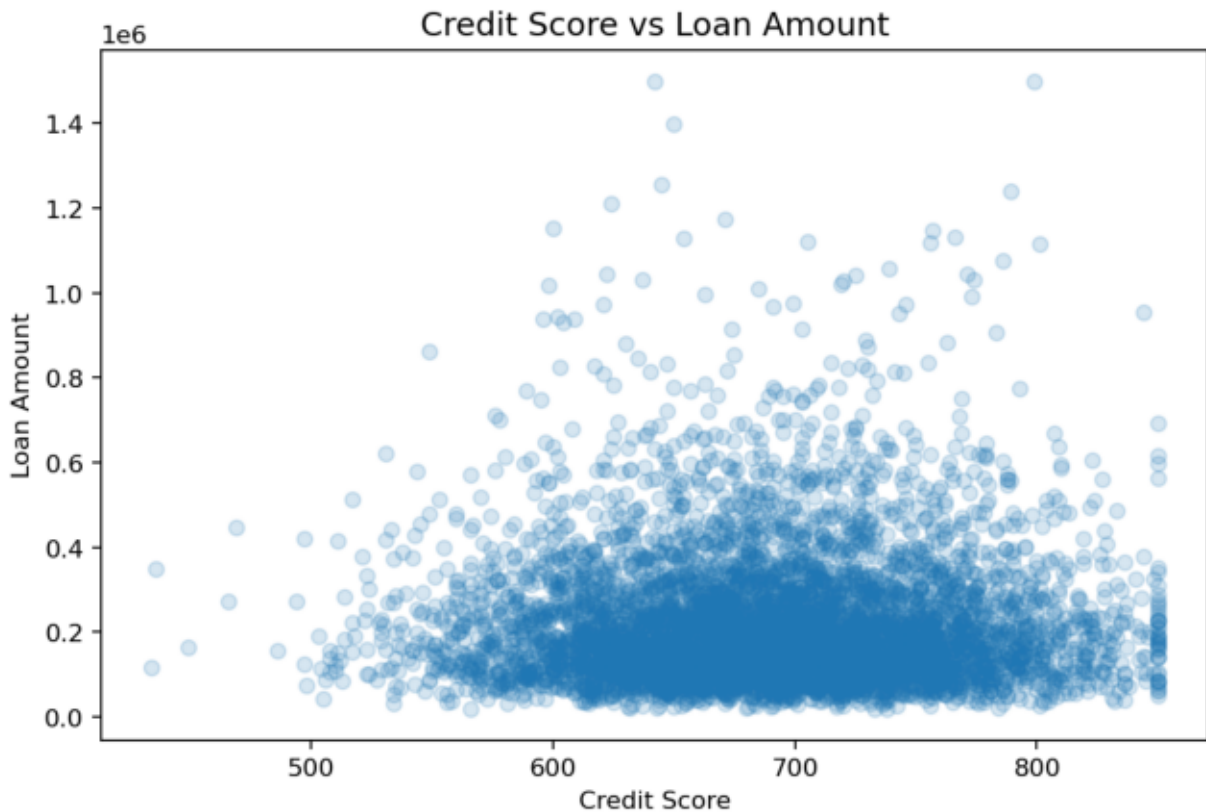
07 Loan Amount Distribution



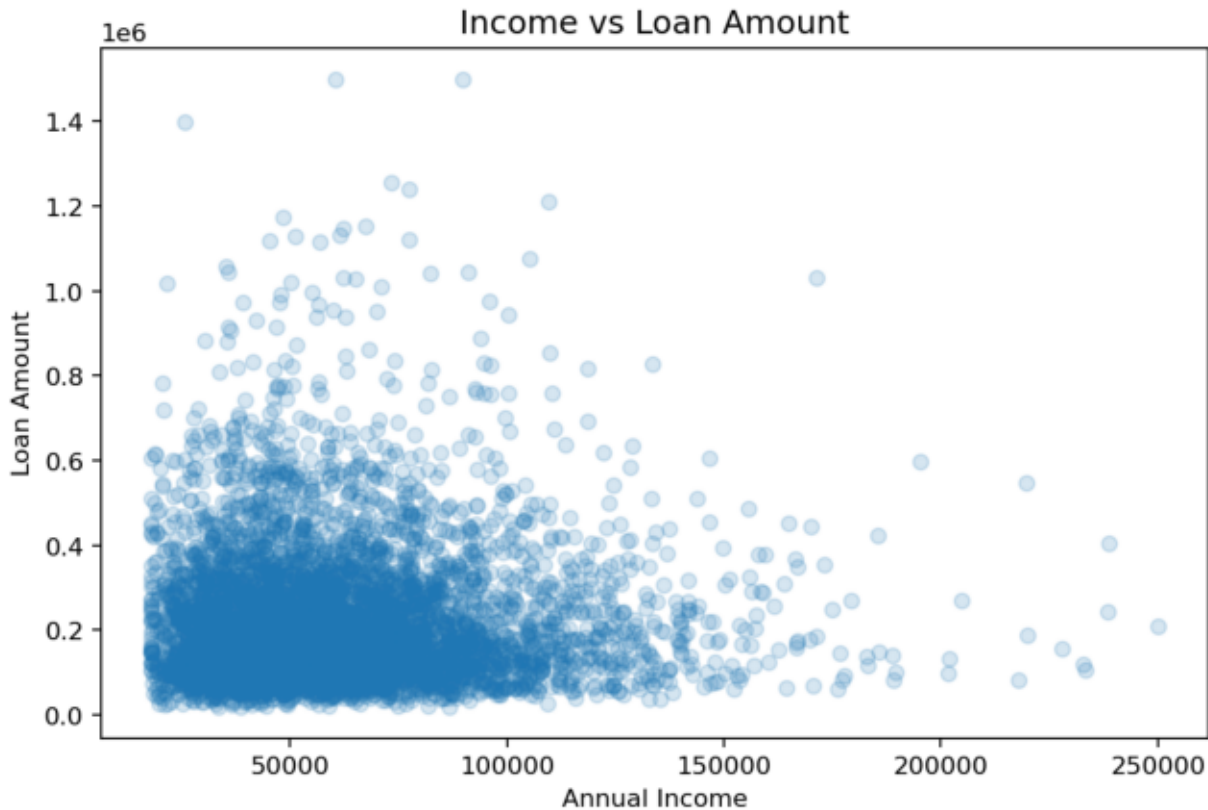
08 Annual Income Distribution



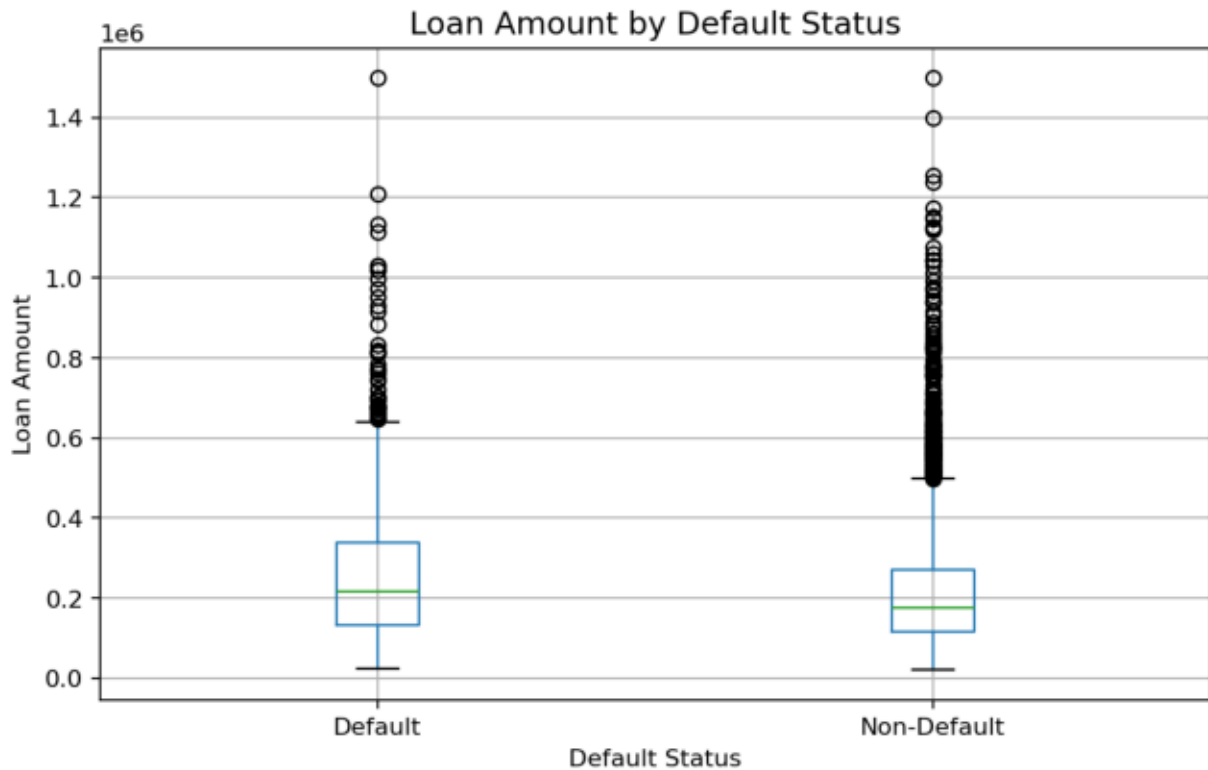
09 Credit Score vs Loan Amount



10 Income vs Loan Amount

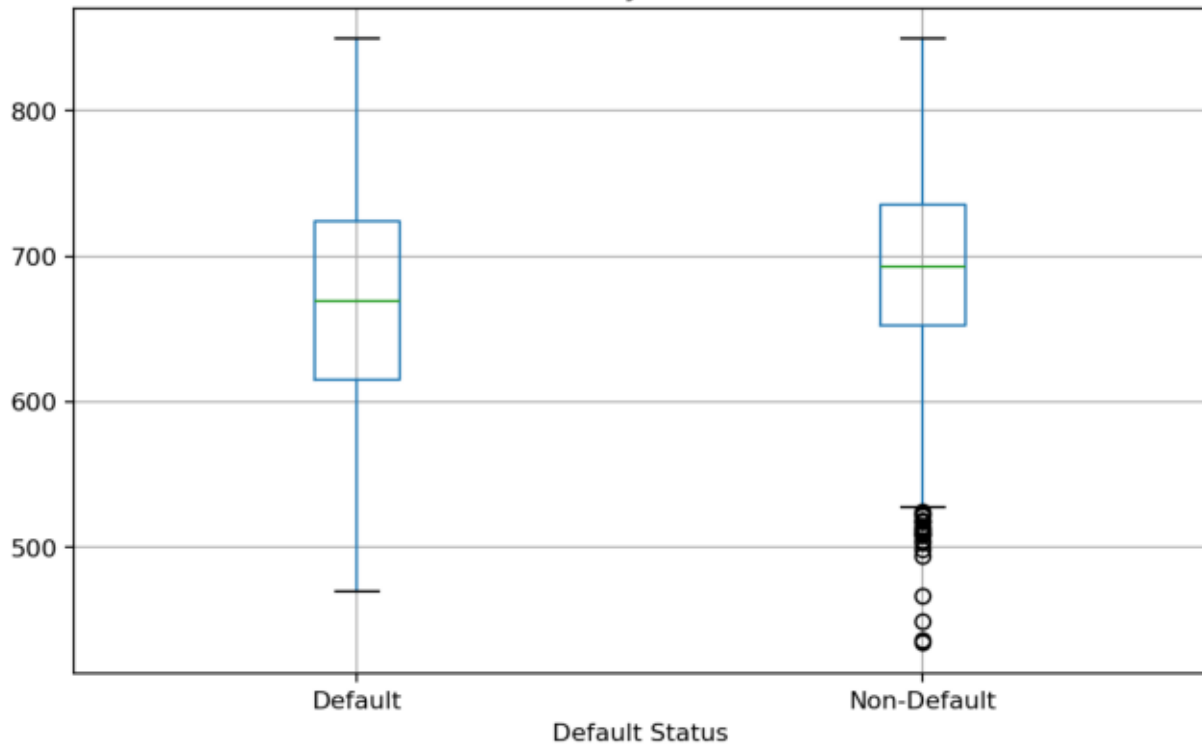


11 Loan Amount by Default

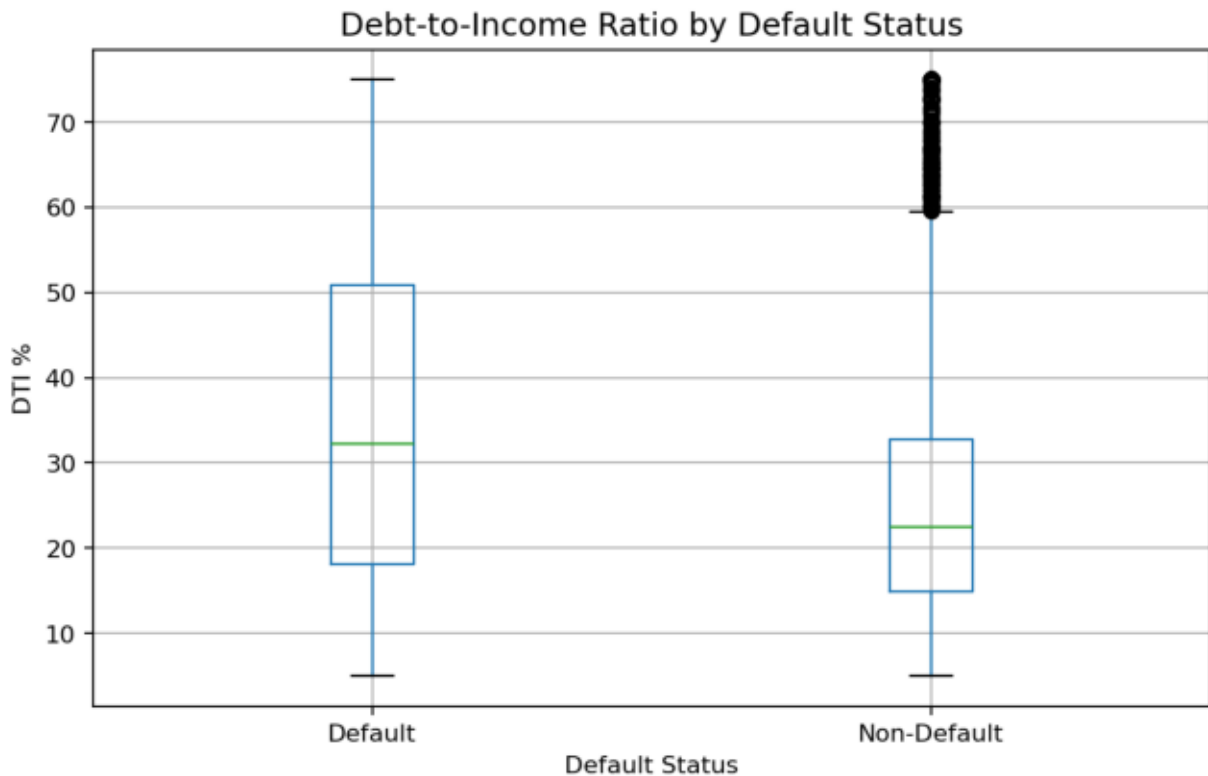


12 Credit Score by Default

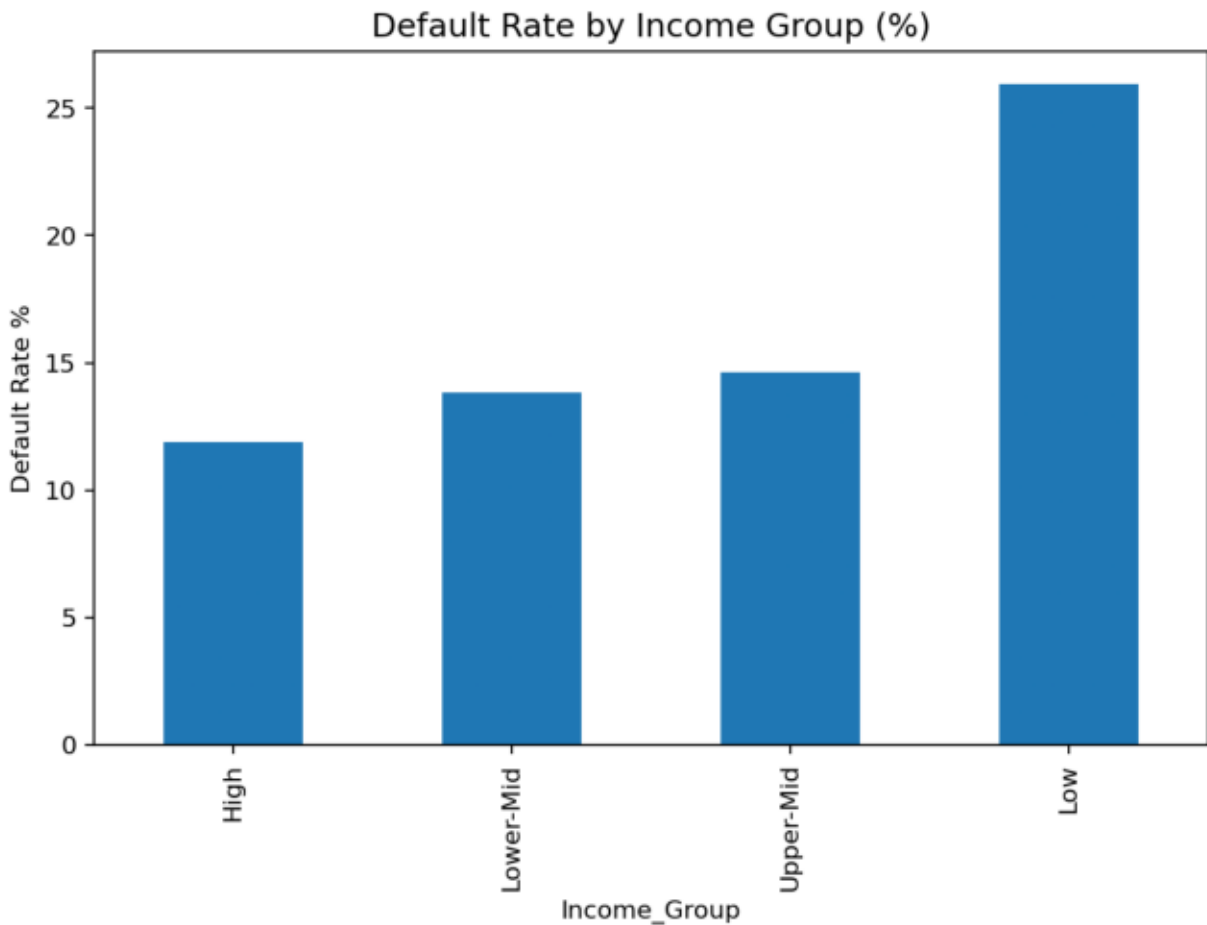
Credit Score by Default Status



13 DTI by Default



14 Default Rate by Income Group



15 Correlation Heatmap

